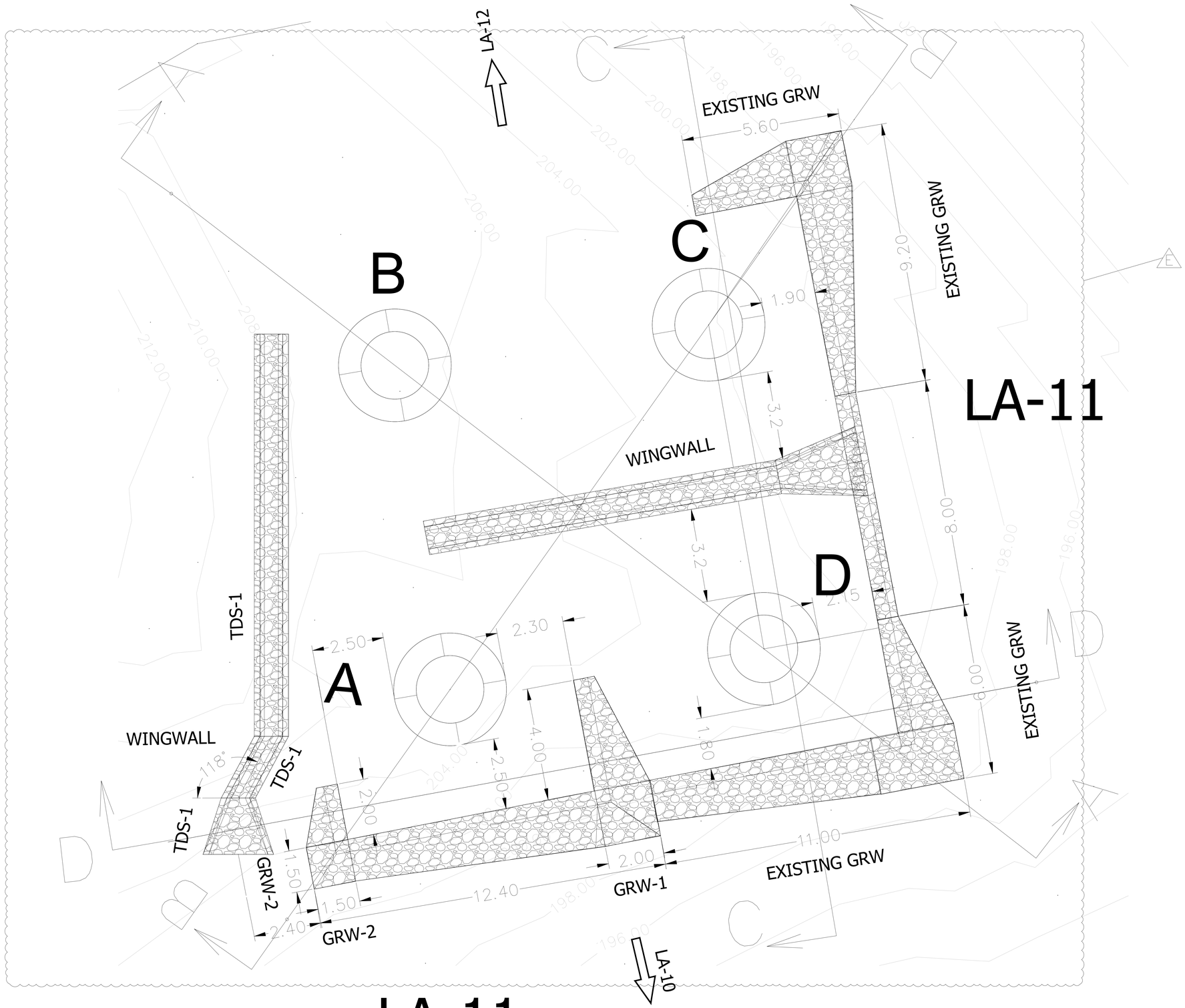
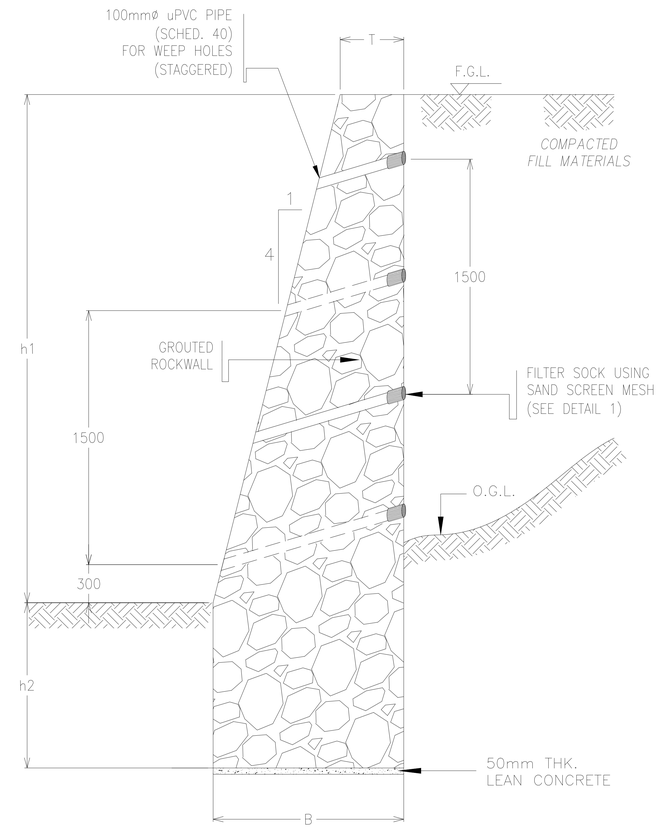



# LA-11

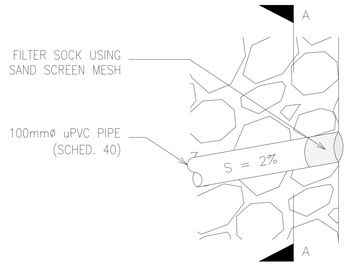
SCALE NTS




## TYPICAL SECTION

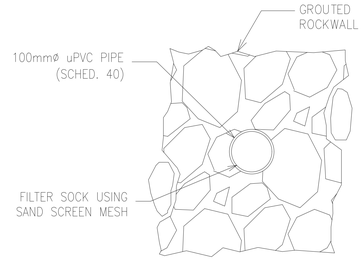
A — SCALE NTS

| TOWER NUMBER | MARK | DIMENSIONS (mm) | | | |
|---|---|---|---|---|---|
| | | h1 | h2 | T | B |
| LA-11 | GRW-1 | 6000 | 3000 | 500 | 2000 |
| | GRW-2 | 4000 | 2000 | 500 | 1500 |
| | VARYING | 1000≤h1 <6000 | h1/2 | 500 | 500+h1/4 |




## DETAIL 1

C — SCALE NTS




## SECTION A-A

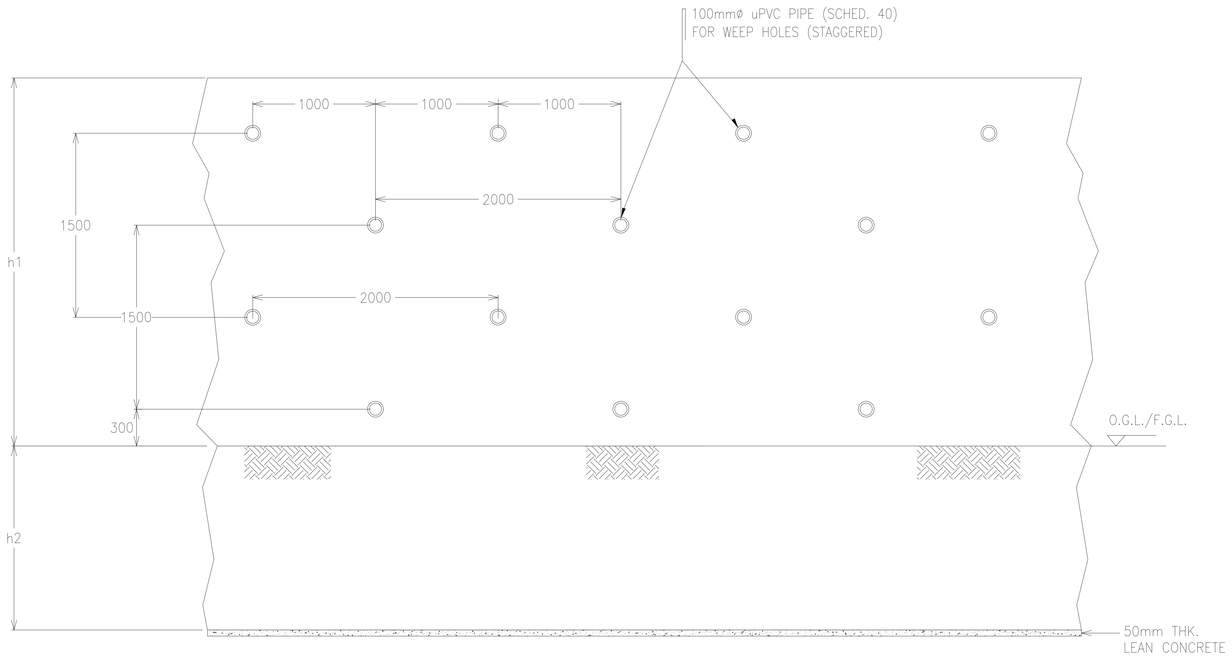
D — SCALE NTS

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS (mm) UNLESS OTHERWISE INDICATED.
2. All VOIDS IN ROCKWALL SHALL BE FILLED WITH GROUT. GROUT SHALL HAVE A MIXTURE OF 1 PART CEMENT TO 3 PARTS SAND.
3. THE LENGTH OF ROCKWALL MAY BE CHANGED TO SUIT ACTUAL FIELD CONDITIONS.
4. REMOVE All BRANCHES, LEAVES, GRASS, ROOTS AND ORGANIC MATERIALS THAT WILL DECOMPOSE BEFORE PLACING FILL MATERIALS AND ROCKWALL STONES.
5. REFER TO SLOPE PROTECTION LAYOUT, FOR THE LOCATION OF GROUTED ROCKWALL.
6. ALL uPVC PIPES SHALL MEET THE REQUIREMENTS OF ASTM DI784.
7. STONES SHALL CONSIST OF HARD,DURABLE MATERIAL, ROUGHLY HAVE THICKNESS OF >15cm, WIDTH OF >1-1.5x ITS THICKNESS & LENGTH OF 1-1.5x ITS WIDTH AND FREE FROM SEAMS, WEATHERED PARTS, DIRTS OR OTHER INJURIOUS MATERIAL.
8. COMPACTED FILL MATERIALS COULD BE A COMBINATION OF EARTH & ROCK PARTICLES NOT GREATER THAN 8 CENTIMETERS OR SANDY CLAY, GRAVELLY CLAY OR SHALE, ALL APPROVED BY THE OWNER.
9. ALL ASPECTS OF CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF NGCP SPECIFICATIONS AND LATEST ACI CODE;
10. THE DESIGN NET ALLOWABLE SOIL BEARING CAPACITY USED WAS TAKEN FROM GEOTECHNICAL REPORTS (MEMO-2021-03-2852 & LETR-2023-03-0311).
11. THE DESIGNER USED THE FOLLOWING SPECIFICATIONS ON THE CALCULATION OF THE PROPOSED SLOPE PROTECTION:
    - UNIT WEIGHT OF WALL: 26.40 kN/m3
    - MATERIAL OF STRUCTURE: STONE MASONRY
    - MASONRY fb = 2.00 MPa
    - Fm = 2.50 MPa
    - COMPRESSIVE STRENGTH: 0.96 MPa
    - SHEAR STRENGTH: 0.10 MPa
    - FLEXURAL TENSILE STRENGTH: 0.05 MPa

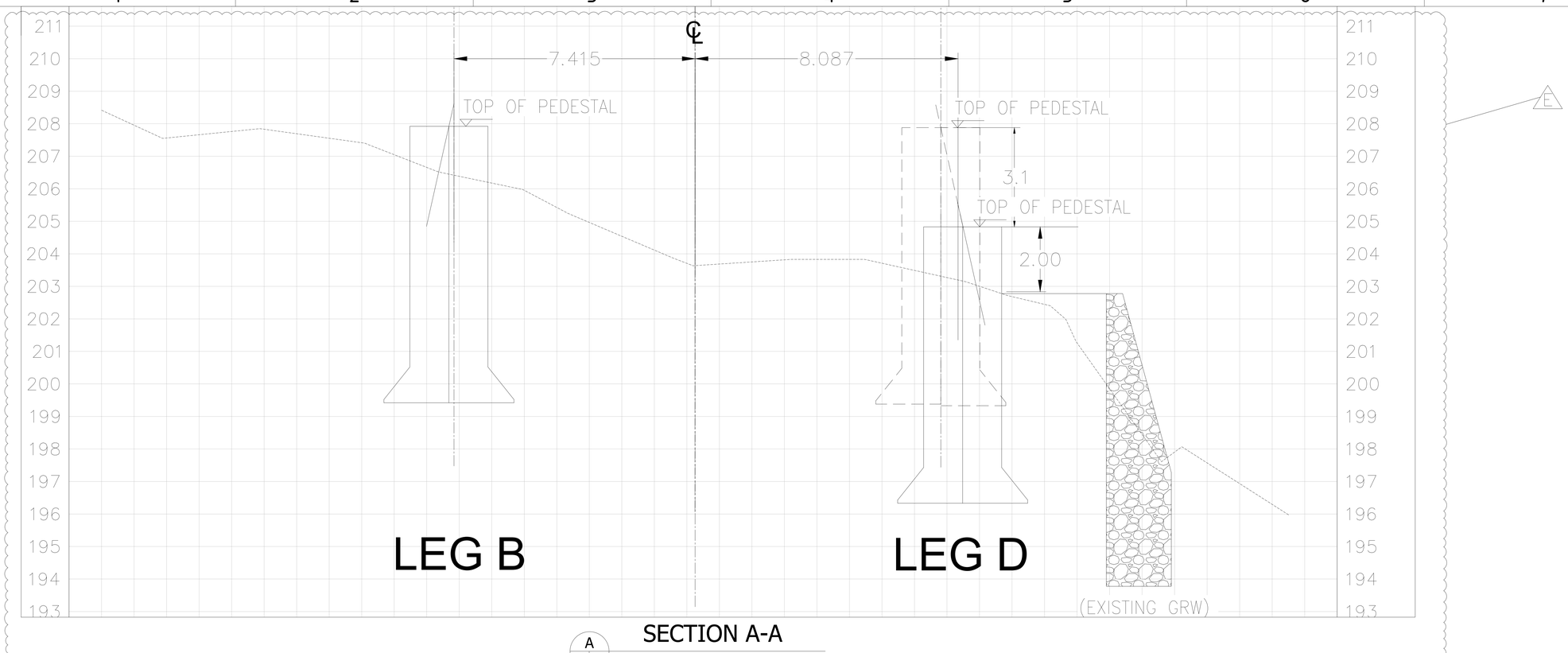
NOMENCLATURES:

| | | |
|---|---|---|
| GRW | – | GROUTED ROCKWALL |
| F.G.L. | – | FINISHED GRADE LINE |
| O.G.L. | – | ORIGINAL GROUND LINE |



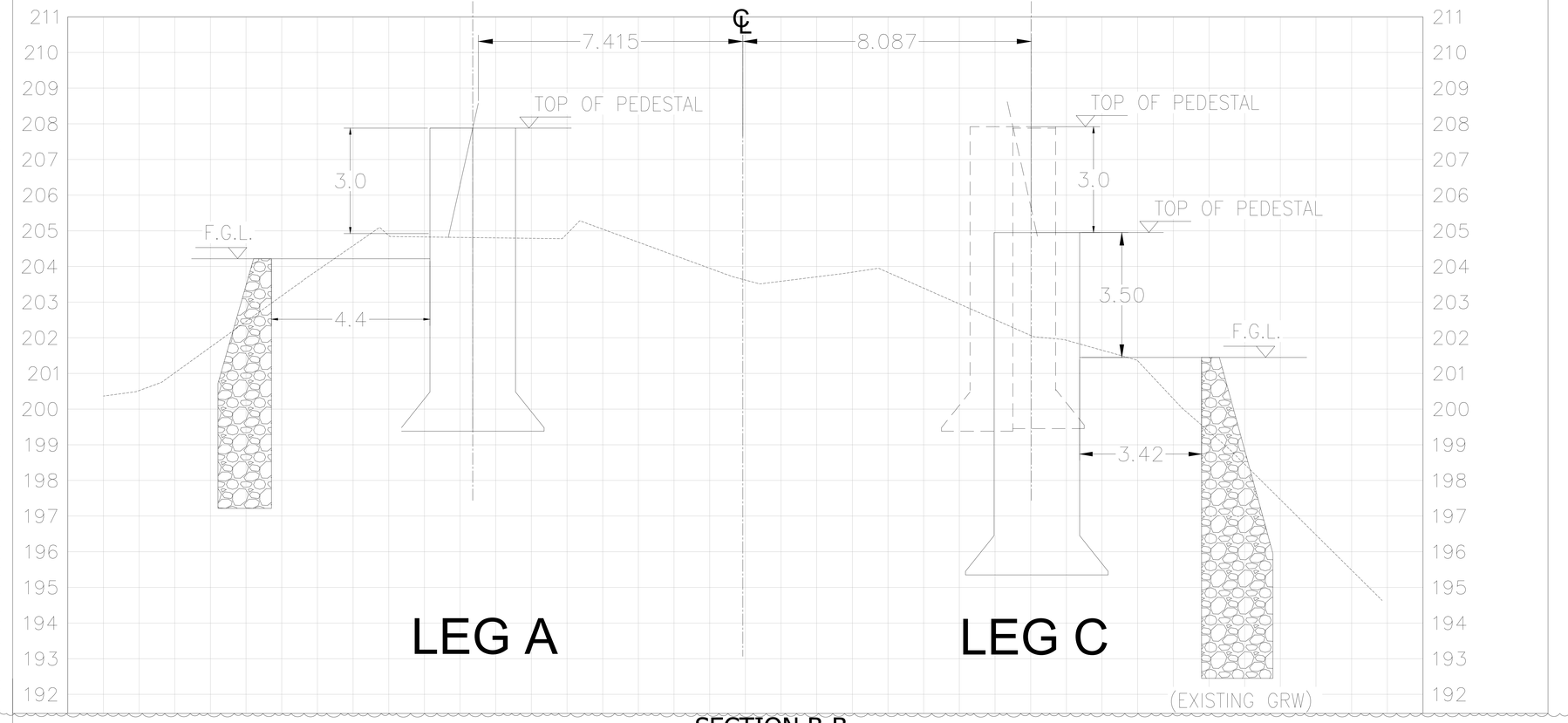

## TYPICAL ELEVATION

B — SCALE NTS



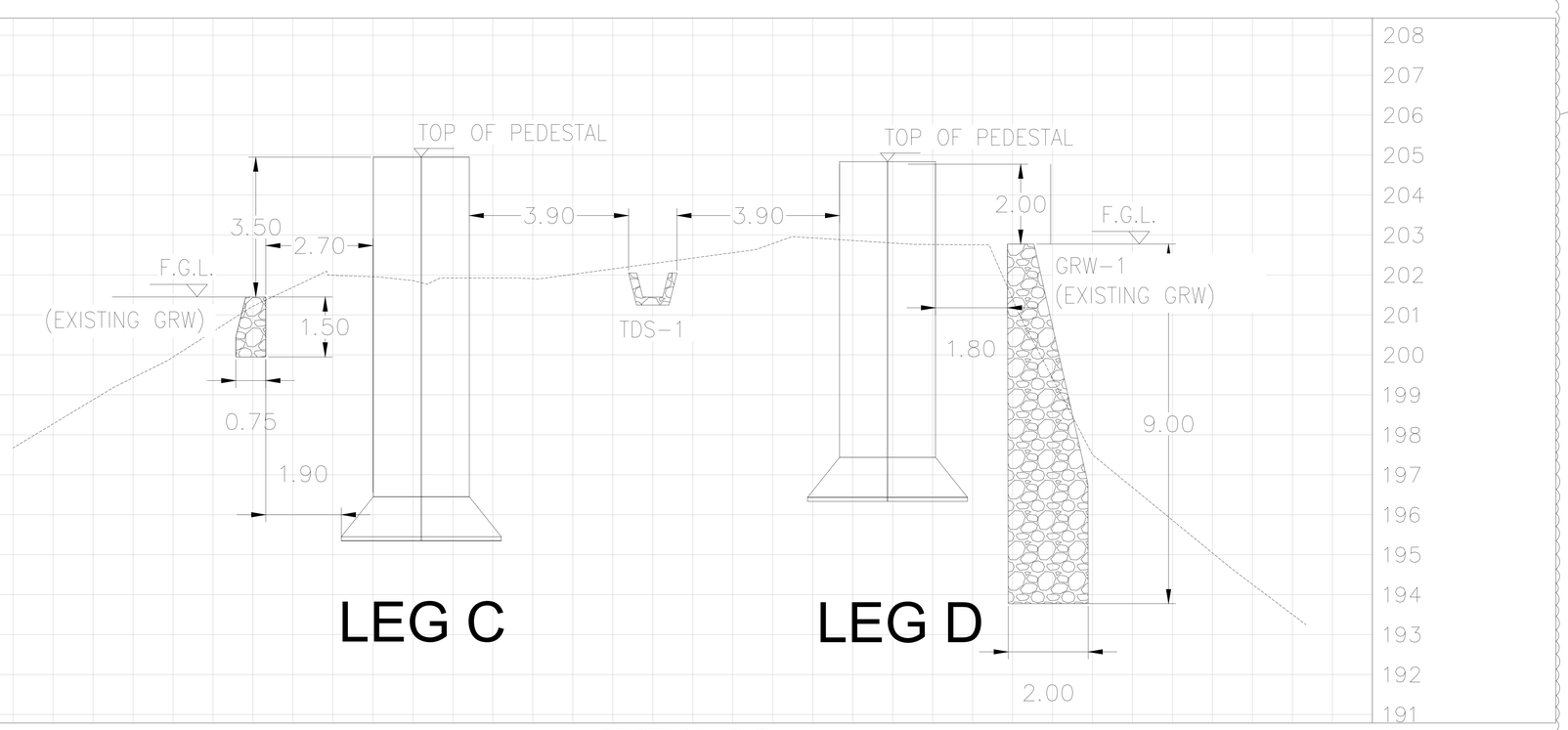

SECTION A-A

SCALE NTS



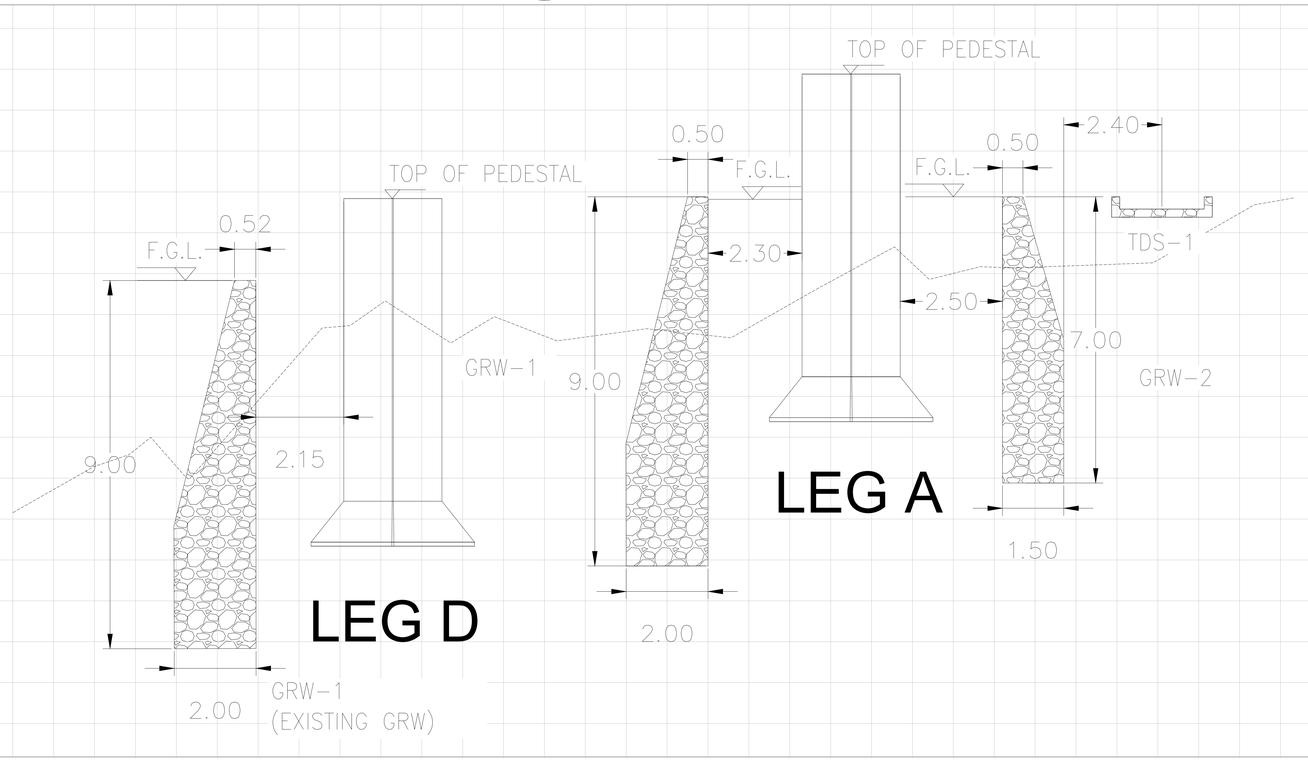

SECTION B-B

SCALE NTS



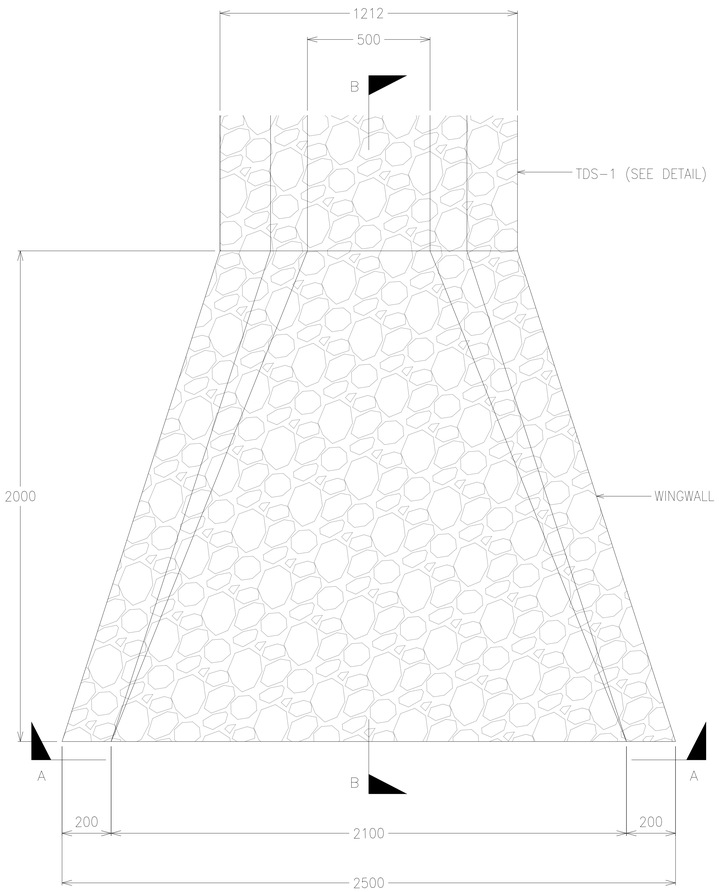

SECTION C-C

SCALE NTS

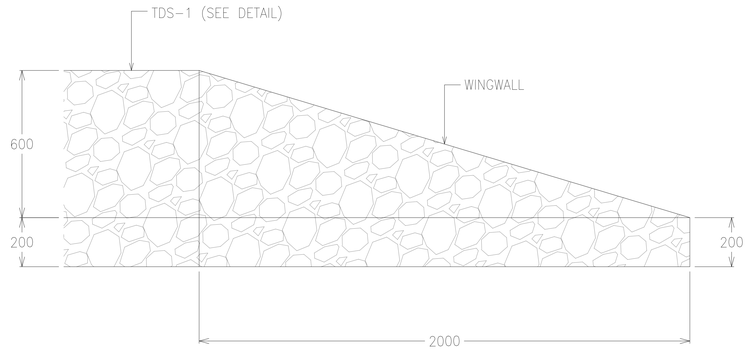



SECTION D-D

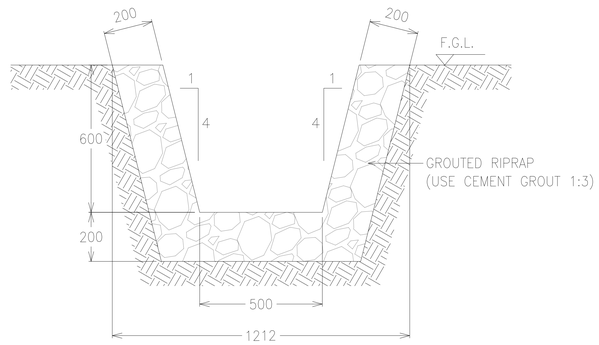
SCALE NTS



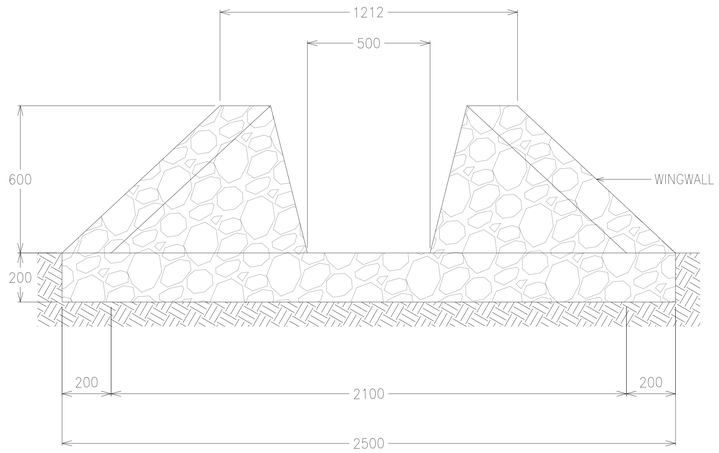

A WINGWALL PLAN
SCALE NTS




C SECTION B
SCALE NTS




D TDS-1 DETAIL
SCALE NTS




B SECTION A
SCALE NTS

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS (mm) UNLESS OTHERWISE INDICATED.
2. COMPACTED FILL MATERIALS COULD BE A COMBINATION OF EARTH & ROCK PARTICLES NOT GREATER THAN 8 CENTIMETERS OR SANDY CLAY, GRAVELLY CLAY OR SHALE, ALL APPROVED BY THE OWNER.
3. ALL ASPECTS OF CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF NGCP SPECIFICATIONS AND LATEST ACI CODE;
4. THE DESIGN NET ALLOWABLE SOIL BEARING CAPACITY USED WAS TAKEN FROM GEOTECHNICAL REPORTS (MEMO-2022-11-0142 & LETR-2023-03-0311).
5. THE DESIGNER USED THE FOLLOWING SPECIFICATIONS ON THE CALCULATION OF THE PROPOSED SLOPE PROTECTION:
   - UNIT WEIGHT OF WALL: 26.40 kN/m3
   - MATERIAL OF STRUCTURE: STONE MASONRY
   - MASONRY fb = 2.00 MPa
   - Fm = 2.50 MPa
   - COMPRESSIVE STRENGTH: 0.96 MPa
   - SHEAR STRENGTH: 0.10 MPa
   - FLEXURAL TENSILE STRENGTH: 0.05 MPa
6. DEPTH AND WIDTH OF CANAL TO BE VERIFIED AT SITE, AND SHALL BE ADJUSTED ACCORDINGLY.
7. FINISHED GRADE ELEVATION SHALL BE SLOPED TO THE DIRECTION TOWARDS THE DRAINAGE OPEN CANAL STRUCTURES.
8. MINIMUM SLOPE USED:
   a. OPEN LINE CANAL = 0.001 (min.)

NOMENCLATURES:

| | | |
|---|---|---|
| F.G.L. | – | FINISHED GRADE LINE |
| O.G.L. | – | ORIGINAL GROUND LINE |
| TDS | – | TRAPEZOIDAL DITCH SECTION |